EXPERIMENT - 7

AIM:

To write a program to implement flow control at data link layer using SLIDING WINDOW PROTOCOL. Simulate the flow of frames from one node to another.

CODE:

SENDER.PY

```
import time
w=int(input("Enter Window Size:"))
msg=input("Enter Message:")
frames=[[i,ch] for i, ch in enumerate(msg)]
base=0
while base<len(frames):
    with open("Sender_Buffer.txt","w") as f:
        for i in range(base, min(base+w, len(frames))):
            f.write(f"{frames[i][0]},
                    {frames[i][1]}\n")
    print("\n Frames Sent:")
    for i in range(base, min(base+w, len(frames))):
        print(frames[i])
    time.sleep(3)
    try:
        with open("Receiver_Buffer.txt","r") as f:
            data=f.read().strip()
        if data.startswith("ACK"):
            ack=int(data.split(":")[1])
            if ack==base+w-1 or ack>=len(frames)-1:
                print("ACK received:", ack)
                base=ack+1
            else:
                print("Wrong ACK Resending....")
```

```
    elif data.startswith("NACK"):
        print("NACK Received. Resending....")

except:
    print("Waiting for Receiver.....")
```

RECEIVER.PY

```
exp = 0
with open("Sender_Buffer.txt","r") as f:
    lines = f.readlines()

ok = True
for line in lines:
    no, data = line.strip().split(",")
    if int(no) != exp:
        ok = false
        nack = exp
        break
    exp+=1

with open("Receiver_Buffer.txt", "w") as f:
    if ok:
        f.write(f"ACK: {exp-1}")
        print("ACK Sent:", exp-1)
    else:
        f.write(f"NACK: {nack}")
        print("NACK Sent:", nack)
```

INPUT:

Enter Window Size: 3

Enter Message: HELLO

Output (Sender)

Frames Sent:

[0, 'H']

[1, 'E']

[2, 'L']

Ack received: 2

Frames sent:

[3, 'L']

[4, 'O']

Ack received: 4

Output (Receiver)

Ack Sent: 2

Ack Sent: 4

RESULT:

Thus the program to implement flow control at data link layer used SLINDING WINDOW has been executed successfully.